

TrES-3b

R-2707

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_110613 · created 2026-09-01T00:50:12+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2455725.90635 +/- 0.00093 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.177 +/- 0.031
Transit depth (Rp/Rs)^2	3.14 +/- 1.11 [%]
Orbital Inclination (inc)	81.81 +/- 1.07
Airmass coefficient 1 (a1)	1.126 +/- 0.016
Airmass coefficient 2 (a2)	-0.095 +/- 0.019
Scatter in the residuals of the lightcurve fit is	1.41 %
Best Comparison Star	None
Optimal Aperture	3.34
Optimal Annulus	11.54
Transit Duration (day)	0.055 +/- 0.013

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.177 ± 0.031 vs 0.16309 (deviation 0.45σ)
Duration fit vs published	0.055 d vs 0.0567 d (deviation 0.13σ)
Mid-time O–C	1.6 min
Depth SNR	5.7
Residual scatter	1.41 %

Light curve

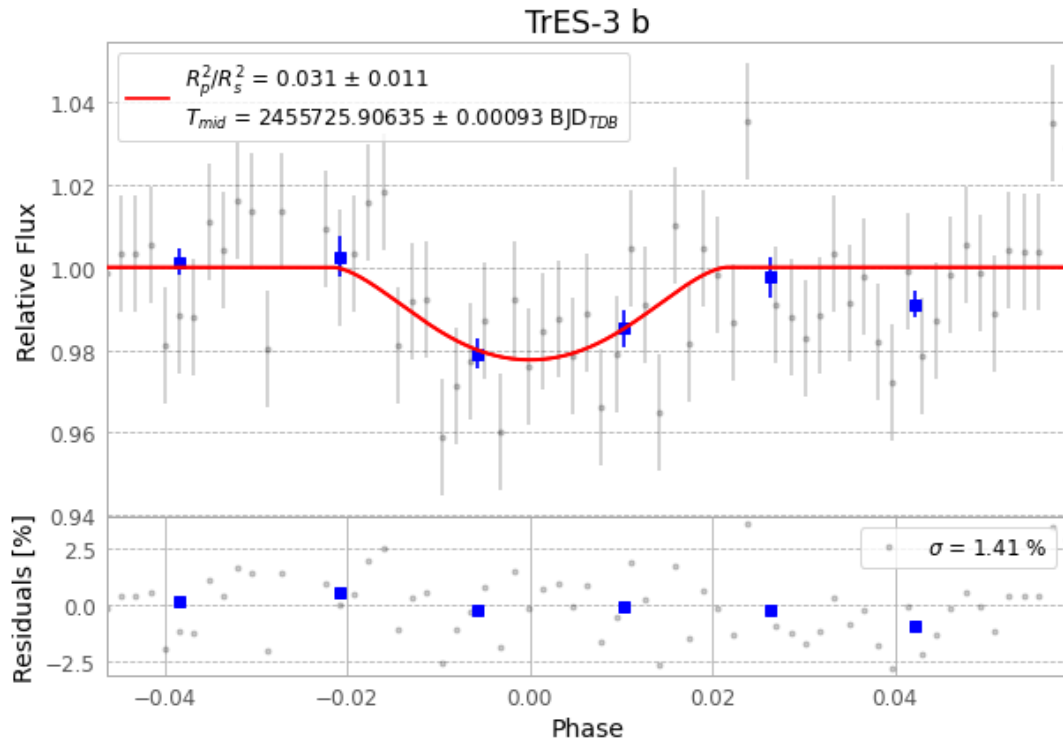


Figure 1 — FinalLightCurve_TrES-3 b_2011-06-13.png (SHA-256 e89ef778e54928fb...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [319, 187], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 7d64dd3a8475781d...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

67 FITS frames (44 MB), TRES-3110613081216.FITS → TRES-3110613113013.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-110613.log (4a9338714703...), FinalLightCurve_TrES-3 b_2011-06-13.png (e89ef778e549...), FinalLightCurve_TrES-3 b_2011-06-13.csv (ae696d563795...)

Compute resources

Wall time 133.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-09-01 00:50Z.